

First Terminal Examination - 2079

Class : XII

Subject: Mathematics (0081)

Full Marks: 75

Time: 3 hrs.

SET 'B'

Pass Marks: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

Group 'A'

Rewrite the best alternatives.

[1 × 11 = 11]

- If $P(n, r) = C(n, r)$ if
 - $r \in \mathbb{N}$
 - $r = 0$ only
 - $r = 1$ only
 - $r = 0$ or 1
- How many two different digit numbers can be formed in which no digit is 5?
 - 64
 - 81
 - 90
 - 100
- In the expansion of $(1 + x)^{50}$, the sum of all binomial coefficients is
 - 0
 - 2^{49}
 - 2^{50}
 - 2^{51}
- The expansion of $\frac{1}{\sqrt{2+5x}}$ is valid if
 - $|5x| < 1$
 - $|x| < \frac{2}{5}$
 - $|x| > \frac{2}{5}$
 - $|x| < \frac{5}{2}$
- The distance between foci of a horizontal ellipse is
 - $\frac{a}{e}$
 - $2ae$
 - $\frac{2a}{e}$
 - ae
- The eccentricity of hyperbola $\frac{x^2}{9} - \frac{y^2}{16} = 1$ is
 - $\frac{4}{5}$
 - $\frac{5}{4}$
 - $\frac{\sqrt{7}}{4}$
 - $\sqrt{2}$

7. The direction cosines of Z-axis are
 a) 1, 0, 0 b) 0, 1, 0 c) 0, 0, 1 d) $\frac{1}{2}, \frac{1}{2}, \frac{1}{\sqrt{2}}$
8. The system of linear equations $2x - y = 1$, $6x - 3y = 3$ are
 a) consistent & dependent b) inconsistent & independent
 c) inconsistent & dependent d) consistent & independent
9. In solving the LP problem using simplex method, if all entries in last row in the simplex table are non-negative then the feasible solution is
 a) minimum b) pivot
 c) optimal d) positive solution
10. The forward elimination step of Gaussian elimination reduces the coefficient matrix of system of equation into
 a) Diagonal matrix b) Identity matrix
 c) Lower triangular matrix d) Upper triangular matrix
11. Differentiability of a function implies
 a) increasing b) decreasing
 c) continuity d) discontinuity

Group 'B'

Short Answer Questions.

[5 × 8 = 40]

12. a) A round table conference is to be held between delegates of 7 countries. In how many ways can they be seated if two particular delegates are
 (i) always together, (ii) never together [2]
- b) A committee of 5 members is to be formed from 6 boys and 4 girls. In how many ways can it be done so as to include at least 2 girls? [3]

13. a) Find the term free from x in the expansion of $\left(2x^2 + \frac{1}{3x}\right)^{12}$. [2]
- b) Find the middle term or terms in the expansion of $\left(\frac{x}{a} - \frac{b}{x}\right)^{2n+1}$. [3]
14. If $(1+x)^n = C_0 + C_1x + C_2x^2 + \dots + C_nx^n$,
prove that:
- i) $C_0 + C_1 + C_2 + \dots + C_n = 2^n$ [1]
- ii) $C_0^2 + C_1^2 + C_2^2 + \dots + C_n^2 = \frac{(2n)!}{n!n!}$ [4]
15. a) Find the vertices, foci and length of latus rectum of the ellipse
 $3x^2 + 4y^2 = 36$ [2]
- b) Find the equation of the hyperbola with foci $(\pm 3, 0)$ and $e = \frac{3}{2}$. [3]
16. a) If α, β & γ be the angles made by a line with the co-ordinate axes,
prove that: $\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 2$. [2]
- b) The direction cosines of two lines are given by $l - 5m + 3n = 0$ and
 $7l^2 + 5m^2 - 3n^2 = 0$. Find the direction cosines of one of the lines. [3]
17. a) Solve using Cramer's rule:
 $3x + \frac{6}{y} = 10$ and $-2x + \frac{3}{y} = -1$ [2]
- b) Solve using row-equivalent matrix method:
 $x + y - z = -3$
 $2x + y + 2z = 16$
 $x + 2y + 3z = 15$ [3]
18. Find from first principle, the derivative of $\ln \sin x$. [5]
19. a) Differentiate $\sin^{-1} \sqrt{1-x^2}$ with respect to x . [2]
- b) If $y = x^y$, prove that: $\frac{dy}{dx} = \frac{y^2}{x(1-y \ln x)}$ [3]

Group 'C'

Long Answer Questions.

[8 × 3 = 24]

20. a) In how many ways can the letters of the word "FRIDAY" be arranged in the following conditions?
- i) If there is no restriction.
 - ii) If vowels are always together.
 - iii) If vowels never come together.
 - iv) If vowels can occupy only even places. [4]
- b) $1 + \frac{1}{3} + \frac{1.3}{3.6} + \frac{1.3.5}{3.6.9} + \dots$ to $\infty = \sqrt{3}$ [4]
21. a) Find the equation of an ellipse whose vertex is at (0, 8) and passes through $(3, \frac{32}{5})$. [3]
- b) Find the direction cosines of a line joining points A(5, 2, 8) and B(7, -1, 9). Also, show that the sum of their squares is 1. [2+1]
- c) If A(3, 4, 5), B(4, 6, 3), C(-1, 2, 4) and D(1, 0, 5) are any four points, find the projection of AB on CD. [2]
22. a) Mention the two steps that are used in Gauss Elimination Method to solve system of linear equations. Use it to solve.
- $$\begin{aligned}x - 3y + z &= 4 \\2x - 8y + 8z &= -2 \\-6x + 3y - 15z &= 9\end{aligned}$$
- [4]
- b) Using simplex method, find the optimal solution of
- $$F = 7x + 5y$$
- subject to
- $$4x + 3y \leq 12$$
- $$x + 2y \leq 6$$
- $$x, y \geq 0$$
- [4]

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